Topological Energy Condensate Theory (TECT): 35

Jusang Lee*
Independent Researcher, Incheon, Korea
(Dated: April, 2026)

I. CANONICAL LOCKED-BASIS READING OF THE LONGITUDINAL ACTUAL PDE COEFFICIENT

II. 1. DIRECT FORMULA FOR $u_\parallel$

Let the shell-relevant reduced J-matrix at the selected intervalley harmonic be

$$\boxed{\mathcal{U}_j(0) = \big(\mathcal{U}_{j,ab}(0)\big)_{a,b=1}^{2}, \qquad \mathcal{U}_{j,ab}(0) = \langle -, a|\widehat{\mathsf{J}}_j(G_*;0)|+, b\rangle.} \tag{1}$$

Then the shell-normal longitudinal singlet coefficient is

$$\boxed{u_\parallel = \frac{1}{2}\,\mathrm{Tr}\big(\Sigma_3\,\hat{n}^j\mathcal{U}_j(0)\big).} \tag{2}$$

In the diagonal Pauli basis $\Sigma_3 = \mathrm{diag}(1,-1)$,

$$\boxed{u_\parallel = \frac{1}{2}\Big[(\hat{n}^j\mathcal{U}_j(0))_{11} - (\hat{n}^j\mathcal{U}_j(0))_{22}\Big].} \tag{3}$$

So $u_\parallel$ is literally the shell-normal diagonal asymmetry of the reduced J-block.

III. 2. DIRECT FORMULA FOR $m_\parallel$

In the canonical locked frame,

$$\boxed{m^a(P_0) = m_\parallel\,\delta^{a3}.} \tag{4}$$

Therefore

$$\boxed{m_\parallel = m^3(P_0).} \tag{5}$$

No further reduction is needed.

IV. 3. THE ACTUAL LONGITUDINAL SCALAR

Hence the unresolved longitudinal scalar is

$$\boxed{L_* = m_\parallel u_\parallel = \frac{m_\parallel}{2}\,\mathrm{Tr}\big(\Sigma_3\,\hat{n}^j\mathcal{U}_j(0)\big).} \tag{6}$$

Equivalently,

$$\boxed{L_* = \frac{m_\parallel}{2}\Big[(\hat{n}^j\mathcal{U}_j(0))_{11} - (\hat{n}^j\mathcal{U}_j(0))_{22}\Big].} \tag{7}$$

* jusanglee@kakao.com

V. 4. DIRECT LONGITUDINAL PDE COEFFICIENT

On the current canonical branch,

$$\boxed{\lambda_{\parallel} = \frac{1}{2}\,\alpha_X\, L_*\, I_{\theta^{03}}.} \tag{8}$$

Comparing with the reference reduced sheet

$$\lambda_{\parallel} = \frac{\lambda_c}{2} I_3,$$

one reads, on this branch,

$$\boxed{\lambda_c = \alpha_X L_*.} \tag{9}$$

Thus the actual longitudinal reduced coefficient is

$$\boxed{\lambda_{\parallel} = \frac{1}{2}\,\alpha_X\, m_{\parallel} u_{\parallel}\, I_{\theta^{03}}.} \tag{10}$$

Using the explicit overlap

$$\boxed{I_{\theta^{03}} = -\frac{3\sqrt{3}\pi}{256}\,\theta_0^3 \kappa^3,} \tag{11}$$

we get

$$\boxed{\lambda_{\parallel} = -\frac{3\sqrt{3}\pi}{512}\,\alpha_X\, m_{\parallel} u_{\parallel}\,\theta_0^3 \kappa^3.} \tag{12}$$

VI. 5. WHAT IS NOW LITERALLY LEFT FOR THE LONGITUDINAL SLOT

The longitudinal slot is therefore no longer structurally open.
The only remaining finite microscopic inputs are

$$\boxed{m_{\parallel} = m^3(P_0), \qquad u_{\parallel} = \frac{1}{2}\,\mathrm{Tr}\big(\Sigma_3\, \hat{n}^j \mathcal{U}_j(0)\big).} \tag{13}$$

Once these two numbers are computed from the actual microscopic branch/operator data, the longitudinal actual PDE coefficient is fully populated.

VII. DIRECT READING OF THE ODD TRANSVERSE COEFFICIENT ON THE ACTIVE ROUTE

The active explicit odd extra-carrier slot is

$$\mathcal{I}_O = X_i^{(O)}\, \cos\theta(\xi)\, J_j D_j \psi.$$

Its induced tangent transport block is

$$V_{EO}^{\mu} = \theta'(\xi)\cos\theta(\xi)\Big[\widetilde{\Lambda}_E\, E^{\mu}{}_a \Sigma_a + \widetilde{\Lambda}_O\, J^{\mu}{}_a \Sigma_a\Big].$$

Hence the odd projector reads

$$\boxed{\widetilde{\Lambda}_O = \frac{\mathsf{P}_O[V_{EO}]}{\theta'(\xi)\cos\theta(\xi)}.}$$

At one-mediator level,

$$\boxed{\widetilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2}\, C_O J_1.}$$

In canonical shell normalization $C_O = 1$,

$$\boxed{\widetilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2}\, J_1.}$$

Therefore the reduced odd transverse coefficient is

$$\boxed{\beta = I_c\, \widetilde{\Lambda}_O = -\frac{\beta_X \gamma_O}{M_X^2}\, C_O J_1\, I_c.}$$

If one uses the transport-side canonical explicit normalization that absorbs J_1,

$$\boxed{\beta = -\frac{\beta_X \gamma_O}{M_X^2}\, I_c.}$$

Under the current reference branch choice

$$\gamma_E = \gamma_O =: \gamma_\perp \neq 0,$$

the reference low-order sheet becomes

$$\boxed{\alpha = \beta = -\frac{\beta_X \gamma_\perp}{M_X^2} I_c.}$$

$$\boxed{\Big[i\partial_t + \frac{3\sqrt{3}\pi}{1024}\, \alpha_X\, m_\parallel\, (u_{11} - u_{22})\, \theta_0^3 \kappa^3\, \sigma_3(-i\partial_\parallel) + \frac{\beta_X \gamma_\perp}{M_X^2} I_c\, (\sigma_1(-i\partial_1) + \sigma_2(-i\partial_2)) + \frac{\beta_X \gamma_\perp}{M_X^2} I_c\, (-\sigma_1(-i\partial_2) + \sigma_2(-i\partial_1))\Big]\chi = 0}$$

VIII. FINITE 2 2 LONGITUDINAL EXTRACTION FOR THE ACTUAL REDUCED PDE

IX. 1. VALLEY-DIAGONAL PROJECTED MATRICES

Fix the explicit carrier basis

$$|+, a\rangle := |G_*, \chi_a\rangle, \qquad |-, a\rangle := |-G_*, \chi_a\rangle, \qquad a = 1, 2.$$

The two valley-diagonal projected blocks at $p = 0$ are

$$\boxed{(E_+(0))_{ab} = \langle +, a|H^{(N)}_{\rm shell}(0)|+, b\rangle, \qquad (E_-(0))_{ab} = \langle -, a|H^{(N)}_{\rm shell}(0)|-, b\rangle.} \tag{14}$$

Equivalently, write them in general Hermitian form

$$\boxed{E_+(0) = \begin{pmatrix} a_+ & c_+ \\ \overline{c_+} & b_+ \end{pmatrix}, \qquad E_-(0) = \begin{pmatrix} a_- & c_- \\ \overline{c_-} & b_- \end{pmatrix},} \tag{15}$$

with

$$a_\pm, b_\pm \in \mathsf{R}, \qquad c_\pm \in \mathsf{C}.$$

X. 2. LOW EIGENVALUES AND NORMALIZED LOW EIGENVECTORS

The lower eigenvalues are

$$\boxed{\varepsilon_{\pm}^{(\mathrm{low})} = \frac{a_{\pm} + b_{\pm}}{2} - \sqrt{\left(\frac{a_{\pm} - b_{\pm}}{2}\right)^2 + |c_{\pm}|^2}.} \tag{16}$$

Define the standard ratio parameter

$$\boxed{r_{\pm} = -\frac{c_{\pm}}{a_{\pm} - \varepsilon_{\pm}^{(\mathrm{low})}}.} \tag{17}$$

Then the normalized low eigenvectors can be written as

$$\boxed{\eta_{\pm} = \frac{1}{\sqrt{1 + |r_{\pm}|^2}} \begin{pmatrix} 1 \\ r_{\pm} \end{pmatrix}.} \tag{18}$$

This is the exact finite 2×2 solution of the valley-side low-mode extraction.

XI. 3. EXACT CLOSED FORMULA FOR Δ_{00}

The intervalley Bloch asymmetry is

$$\boxed{\Delta_{00} = \eta_{+}^{\dagger} \Sigma_3 \eta_{-}.} \tag{19}$$

Substituting (18), one gets the exact closed form

$$\boxed{\Delta_{00} = \frac{1 - \overline{r_{+}} r_{-}}{\sqrt{(1 + |r_{+}|^2)(1 + |r_{-}|^2)}}.} \tag{20}$$

Hence the exact longitudinal cancellation condition is

$$\boxed{\Delta_{00} = 0 \iff \overline{r_{+}} r_{-} = 1.} \tag{21}$$

So the first longitudinal finite audit is reduced completely to the microscopic matrix entries

$$a_{\pm}, \qquad b_{\pm}, \qquad c_{\pm}.$$

XII. 4. REDUCED SHELL-NORMAL INTERVALLEY J-MATRIX

Define the shell-normal intervalley J-matrix

$$\boxed{U_{\parallel} = \hat{n}^j \mathcal{U}_j(0) = \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix}, \qquad (\mathcal{U}_j(0))_{ab} = \langle -, a | \widehat{\mathrm{J}}_j(G_*; 0) | +, b \rangle.} \tag{22}$$

The longitudinal bias is

$$\boxed{u_{\parallel} = \frac{1}{2} \operatorname{Tr}(\Sigma_3 U_{\parallel}) = \frac{1}{2}(u_{11} - u_{22}).} \tag{23}$$

Therefore the actual unresolved longitudinal scalar is

$$\boxed{L_* = m_{\parallel} u_{\parallel} = \frac{m_{\parallel}}{2}(u_{11} - u_{22}).} \tag{24}$$

XIII. 5. FINITE HARMONIC FORMULA FOR $u_\parallel$

In the shell-harmonic basis,

$$u_a^{\pm}(x) = \sum_{G\in\mathcal{S}_N} c_{a,G}^{(\pm)} e^{iG\cdot x}, \qquad \mathsf{J}_j(x) = \sum_Q \widehat{\mathsf{J}}_j(Q;0) e^{iQ\cdot x}.$$

Then

$$\boxed{(\mathcal{U}_j(0))_{ab} = V_{\rm cell} \sum_{G,G^0\in\mathcal{S}_N} \overline{c_{a,G}^{(+)}}\, \widehat{\mathsf{J}}_j(G_* + G - G';0)\, c_{b,G^0}^{(-)}.} \tag{25}$$

Hence

$$\boxed{u_\parallel = \frac{V_{\rm cell}}{2} \sum_{G,G^0\in\mathcal{S}_N} \Big(\overline{c_{1,G}^{(+)}}\, c_{1,G^0}^{(-)} - \overline{c_{2,G}^{(+)}}\, c_{2,G^0}^{(-)}\Big) \hat{n}^j \widehat{\mathsf{J}}_j(G_* + G - G';0).} \tag{26}$$

Equivalently,

$$\boxed{L_* = \frac{m_\parallel V_{\rm cell}}{2} \sum_{G,G^0\in\mathcal{S}_N} \Big(\overline{c_{1,G}^{(+)}}\, c_{1,G^0}^{(-)} - \overline{c_{2,G}^{(+)}}\, c_{2,G^0}^{(-)}\Big) \hat{n}^j \widehat{\mathsf{J}}_j(G_* + G - G';0).} \tag{27}$$

XIV. 6. LONGITUDINAL COEFFICIENT OF THE ACTUAL REDUCED PDE

On the current canonical branch,

$$\boxed{\lambda_\parallel = \frac{1}{2}\,\alpha_X\, L_*\, I_{\theta^{03}}.} \tag{28}$$

Using the explicit overlap

$$\boxed{I_{\theta^{03}} = -\frac{3\sqrt{3}\pi}{256}\,\theta_0^3\kappa^3,} \tag{29}$$

we obtain

$$\boxed{\lambda_\parallel = -\frac{3\sqrt{3}\pi}{512}\,\alpha_X\, L_*\,\theta_0^3\kappa^3 = -\frac{3\sqrt{3}\pi}{1024}\,\alpha_X\, m_\parallel\,(u_{11}-u_{22})\,\theta_0^3\kappa^3.} \tag{30}$$

XV. 7. ACTUAL REDUCED PDE COEFFICIENT SHEET

The active reference-route reduced PDE coefficient sheet is therefore

$$\boxed{\lambda_\parallel = -\frac{3\sqrt{3}\pi}{1024}\,\alpha_X\, m_\parallel\,(u_{11}-u_{22})\,\theta_0^3\kappa^3,} \tag{31}$$

$$\boxed{\alpha = \beta = -\frac{\beta_X\gamma_\perp}{M_X^2}\, I_c.} \tag{32}$$

Thus the current reference reduced PDE is

$$\boxed{\Big[i\partial_t + \frac{3\sqrt{3}\pi}{1024}\,\alpha_X\, m_\parallel\,(u_{11}-u_{22})\,\theta_0^3\kappa^3\,\sigma_3(-i\partial_\parallel) + \frac{\beta_X\gamma_\perp}{M_X^2} I_c\,\big(\sigma_1(-i\partial_1)+\sigma_2(-i\partial_2)\big) + \frac{\beta_X\gamma_\perp}{M_X^2} I_c\,\big(-\sigma_1(-i\partial_2)+\sigma_2(-i\partial_1)\big)\Big]\chi = 0.} \tag{33}$$

The first sign is positive because the PDE contains $-\lambda_\parallel$, while $\lambda_\parallel$ itself is negative in (30).

XVI. 8. WHAT REMAINS LITERALLY TO EVALUATE

No structural ambiguity remains in the longitudinal slot.
The remaining microscopic inputs are exactly

$$\boxed{a\ ;\quad b\ ;\quad c\ ;\quad u_{11};\quad u_{22};\quad m_k:} \tag{34}$$

Equivalently, at the operator level, the true remaining frontier is to evaluate

$$\boxed{c^{(\ )}_{a;G}\ ;\qquad \mathfrak{h}_j\,(Q;0);\qquad P_0;\qquad m^3(P_0)}$$

for the nal re ned locked microscopic operator.

XVII. FINITE 2 2 LONGITUDINAL INSERTION SHEET

XVIII. 1. VALLEY-DIAGONAL PROJECTED MATRICES

Write

$$E_+(0) = \begin{matrix} a_+ & c_+ \\ \overline{c_+} & b_+ \end{matrix}\ ;\qquad E\ \ (0) = \begin{matrix} a & c \\ \overline{c} & b \end{matrix}\ ;$$

with

$$a\ ; b\ \ 2\ \mathbb{R};\qquad c\ \ 2\ \mathbb{C}:$$

Their low eigenvalues are

$$"_{(\mathrm{low})} = \frac{a\ +b}{2}\ \ s\ \frac{}{\left(\frac{a\ \ b}{2}\right)^2 + jc\ \ j^2}:$$

De ne

$$r\ \ = \frac{c}{a\ \ "_{(\mathrm{low})}} = \frac{r\ \ \dfrac{c}{\left(\frac{a\ \ b}{2}\right)^2 + jc\ \ j^2}}{\frac{a\ \ b}{2} + \sqrt{\left(\frac{a\ \ b}{2}\right)^2 + jc\ \ j^2}}:$$

Then the normalized low eigenvectors can be chosen as

$$= p\frac{1}{1 + jr\ \ j^2}\begin{matrix} 1 \\ r \end{matrix}\ :$$

XIX. 2. EXACT FINITE FORMULA FOR $_{00}$

$$_{00} = \begin{matrix} y \\ + \end{matrix}\ _3\ ;\qquad _3 = \mathrm{diag}(1;\ 1):$$

Hence

$$\boxed{_{00} = p\frac{1\ \ \overline{r_+}r}{(1 + jr_+ j^2)(1 + jr\ \ j^2)}:}$$

Therefore

$$\boxed{_{00} = 0\ \ (\ )\qquad \overline{r_+}r\ \ = 1:}$$

XX. 3. SHELL-NORMAL REDUCED J-MATRIX

De ne

$$U_k := \hat{n}^j U_j(0) = \begin{matrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{matrix}; \qquad (U_j(0))_{ab} = h\,; a\,j\,\hat{b}_j(G\;;0)j+;bi:$$

Then the longitudinal J-singlet bias is

$$\boxed{u_k = \frac{1}{2}\mathrm{Tr}(\quad {}_3U_k) = \frac{1}{2}(u_{11} \quad u_{22}):}$$

Hence

$$\boxed{L \quad = m_k u_k = \frac{m_k}{2}(u_{11} \quad u_{22}):}$$

XXI. 4. ACTUAL LONGITUDINAL COEFFICIENT

On the current canonical branch,

$$_k = \frac{1}{2} \quad \chi \; L \quad I_{03}:$$

Using

$$I_{03} = \frac{3\sqrt{3}}{256} \; {}^{3}_{0} \; {}^{3};$$

we obtain

$$\boxed{_k = \frac{3\sqrt{3}}{1024} \quad \chi \; m_k(u_{11} \quad u_{22}) \; {}^{3}_{0} \; {}^{3}:}$$

XXII. 5. CURRENT ACTUAL REDUCED PDE COEFFICIENT SHEET

$$\boxed{_k = \frac{3\sqrt{3}}{1024} \quad \chi \; m_k(u_{11} \quad u_{22}) \; {}^{3}_{0} \; {}^{3}; \qquad = \; = \; \frac{\chi \; ?}{M_{\chi}^{2}} I_c:}$$

XXIII. 6. LITERAL REMAINING MICROSCOPIC DATA

$$\boxed{a \;; \quad b \;; \quad c \;; \quad u_{11}; \quad u_{22}; \quad m_k:}$$

Equivalently:

$$\boxed{E \quad (0); \qquad U_k = \hat{n}^j U_j(0); \qquad m_k = m^{\,3}(P_0):}$$

XXIV. FINAL FINITE INSERTION SHEET FOR THE ACTUAL REDUCED PDE

XXV. 1. VALLEY-SIDE 2 2 PROJECTED BLOCKS

Write

$$E_+(0) = \begin{matrix} a_+ & c_+ \\ \overline{c_+} & b_+ \end{matrix} ; \qquad E_-(0) = \begin{matrix} a_- & c_- \\ \overline{c_-} & b_- \end{matrix} ;$$

with

$$a_\pm ; b_\pm \in \mathbb{R}; \qquad c_\pm \in \mathbb{C}:$$

Their lower eigenvalues are

$$\varepsilon_{\pm}^{(\mathrm{low})} = \frac{a_\pm + b_\pm}{2} - \sqrt{\left(\frac{a_\pm - b_\pm}{2}\right)^2 + |c_\pm|^2}:$$

De ne

$$r_\pm = \frac{c_\pm}{a_\pm - \varepsilon_{\pm}^{(\mathrm{low})}}; \qquad \tau_\pm = e^{i\theta_\pm}\frac{1}{\sqrt{1+|r_\pm|^2}}\begin{matrix} 1 \\ r_\pm \end{matrix} :$$

Then the exact intervalley asymmetry is

$$\boxed{\Delta_{00} = \tau_+^{\dagger}\sigma_3\tau_- = e^{i\theta_+}\frac{1 - \overline{r_+}\,r_-}{(1+|r_+|^2)(1+|r_-|^2)}:}$$

XXVI. 2. SHELL-NORMAL REDUCED J-MATRIX

De ne

$$U_k := \hat{n}^j U_j(0) = \begin{matrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{matrix} ; \qquad (U_j(0))_{ab} = \langle a; \alpha_j b\rangle + \langle a; \beta_j(G_\perp; 0)|+; b\rangle:$$

Then

$$\boxed{u_k = \frac{1}{2}\mathrm{Tr}(\sigma_3 U_k) = \frac{1}{2}(u_{11} - u_{22}):}$$

Equivalently, in the shell-harmonic basis,

$$(U_j(0))_{ab} = V_{\mathrm{cell}} \sum_{G;G' \in S_N} \overline{c^{(+)}_{a;G}}\, \beta_j(G_\perp + G_\perp - G'_\perp, 0)\, c^{(-)}_{b;G'}:$$

Hence

$$\boxed{u_k = \frac{V_{\mathrm{cell}}}{2} \sum_{G;G' \in S_N} \left(\overline{c^{(+)}_{1;G}}\, c^{(-)}_{1;G'} - \overline{c^{(+)}_{2;G}}\, c^{(-)}_{2;G'}\right) \hat{n}^j \beta_j(G_\perp + G_\perp - G'_\perp, 0):}$$

XXVII. 3. LONGITUDINAL SCALAR AND COEFFICIENT

Let

$$m_k = m^{\parallel}(P_0):$$

Then

$$\boxed{L \;= m_k u_k = \frac{m_k}{2}(u_{11} \;\; u_{22}):}$$

Using

$$I_{03} = \frac{3\sqrt{3}}{256} \;\; {}^{3}_{0} \;\; {}^{3};$$

the actual longitudinal reduced coe cient is

$$\boxed{{}_k = \frac{1}{2} \;\; {}_X L \;\; I_{03} = \frac{3\sqrt{3}}{1024} \;\; {}_X \; m_k (u_{11} \;\; u_{22}) \;\; {}^{3}_{0} \;\; {}^{3}:}$$

XXVIII. 4. TRANSVERSE COEFFICIENTS ON THE ACTIVE REFERENCE ROUTE

The overlap dictionary is

$$I_c = \int d \; f_+(\;)f \quad (\;) \quad {}^{0}(\;)\cos(\;):$$

On the active explicit odd extra-carrier route,

$${}_k = \frac{c}{2} I_3; \qquad = I_{\;c} e_E; \qquad = I_{\;c} e_O:$$

At one-mediator level in canonical explicit normalization,

$$e_E^{(1)} = \frac{{}_X \;\; E}{M_X^2}; \qquad e_O^{(1)} = \frac{{}_X \;\; O}{M_X^2}:$$

Under the current reference choice

$${}_E = \;\; {}_O = \;\; {}_?;$$

this becomes

$$\boxed{= \;\; = \;\; \frac{{}_X \;\; ?}{M_X^2} I_c:}$$

XXIX. 5. FINAL CURRENT REFERENCE REDUCED PDE

Therefore the current actual/reference reduced PDE is

$$\boxed{\left[i@_t + \frac{3\sqrt{3}}{1024} \;\; {}_X \; m_k (u_{11} \;\; u_{22}) \;\; {}^{3}_{0} \;\; {}^{3} \;\; {}_3(\,i@_k) + \frac{{}_X \;\; ?}{M_X^2} I_c \;\; {}_1(\,i@_1) + \;\; {}_2(\,i@_2) \;\; + \frac{{}_X \;\; ?}{M_X^2} I_c \;\; {}_1(\,i@_2) + \;\; {}_2(\,i@_1) \right]}$$

XXX. 6. LITERAL REMAINING MICROSCOPIC INPUTS

So the only remaining microscopic data to insert are

$$\boxed{a \;\; ; \;\; b \;\; ; \;\; c \;\; ; \;\; u_{11}; \;\; u_{22}; \;\; m_k; \;\; I_c:}$$

Equivalently:

$$\boxed{E \;\; (0); \qquad U_k; \qquad m^3(P_0); \qquad I_c:}$$

Once these are evaluated from the actual microscopic branch/operator data, the coe cient point

$$(\ _k; \ ; \ )$$

is fully populated.

$$\boxed{h \quad i@_q + \frac{p\overline{3}}{512} \ X \ {}^{3}_{0} \ {}^{3} \ _3(i@_k) + \frac{p\overline{3}}{16} \ X \ _0 \quad _1(i@_1) + \ _2(i@_2) \ + \frac{p\overline{3}}{16} \ X \ _0 \quad _1(i@_2) + \ _2(i@_1) \quad i = 0}$$

$$\boxed{h \quad i@_q + \frac{p\overline{3}}{512} \ X \ {}^{3}_{0} \ {}^{3} \ _3(i@_k) + \frac{p\overline{6}}{16} \ X \ _0 \quad _1(i@_{q_1}) + \ _2(i@_{q_2}) \quad i = 0}$$

XXXI. FINAL FULL-MICROSCOPIC EXTRACTION ROUTE

XXXII. 1. SHELL-PROJECTED FINITE OPERATOR

Choose the folded Bragg point

$$k \ = \frac{G}{2}; \qquad k = k \ + p; \qquad jpj \ \ jG \ \ j:$$

Let S be the nite near-resonant harmonic set around k , and de ne the shell projector

$$_S \ : H_{Bloch}(k) \ ! H \ _S(k):$$

Then the rst nite-dimensional operator belonging to the actual full-PDE derivation is

$$\boxed{H_{shell}(p) = \ _S \ H_e \ (k \ + p) \ _S \ :}$$

This is the true computational bottleneck. :contentReference[oaicite:2]index=2

XXXIII. 2. LOW VALLEY PROJECTION

Assume $H_{shell}(0)$ contains an isolated two-dimensional low-energy sector V. Let

$$\boxed{P \ : H_S \ (0) \ ! V}$$

be the spectral projector onto that valley subspace.

De ne the rst derivatives

$$\boxed{K_i \ := \ @_{p_i} H_{shell}(p)j_{p=0} \ :}$$

Then the projected linear valley operator is

$$\boxed{H^{(1)}_{val}(p) = p_i \ P \ K_i P \ :}$$

:contentReference[oaicite:3]index=3

XXXIV. 3. UNIFIED COEFFICIENT READOUT

Inside the valley space, de ne

$$M_i = \frac{1}{2}\mathrm{Tr}_H \quad P\ K_i P \quad ; \qquad = 1;2;3:$$

In the shell-adapted frame $(\hat{n};e_1;e_2)$,

$$_k = \hat{n}_i M_{3i};$$

$$= \frac{1}{2}\ e_{1i}M_{1i} + e_{2i}M_{2i}\ ; \qquad = \frac{1}{2}\ e_{2i}M_{1i} \quad e_{1i}M_{2i}\ :$$

Equivalently,

$$_k = \frac{1}{2}\mathrm{Tr}(\ _3P\ K_k P\ ); \qquad K_k = \hat{n}_i K_i :$$

:contentReference[oaicite:4]index=4 :contentReference[oaicite:5]index=5

XXXV. 4. MINIMAL AXIAL 4 4 SHELL ANSATZ

On the folded-Bragg 12-mode shell pair, the minimal axial ansatz is

$$K_i = A\hat{n}_i \ ^3\ 1\ _2 + B\,(e_{1i}\ ^1\ s\ _1 + e_{2i}\ ^1\ s\ _2) + C\,(e\ _{2i}\ ^1\ s\ _1 + e_{1i}\ ^1\ s\ _2):$$

Hence

$$K_k = A\ ^3\ 1\ _2;$$

$$K_1 = B\ ^1\ s\ _1 + C\ ^1\ s\ _2; \qquad K_2 = C\ ^1\ s\ _1 + B\ ^1\ s\ _2:$$

Projecting to the selected positive-valley two-band block,

$$P_+(\ ^3\ 1\ _2)P_+ \ 7!\ ^3; \qquad P_+(\ ^1\ s\ _1)P_+ \ 7!\ ^1; \qquad P_+(\ ^1\ s\ _2)P_+ \ 7!\ ^2;$$

so that

$$h_+(p) = A\,p_k\ ^3 + B(p_1\ ^1 + p_2\ ^2) + C(p\ _2\ ^1 + p_1\ ^2) + O(jpj\ ^2):$$

Therefore

$$_k = A; \qquad = B; \qquad = C:$$

:contentReference[oaicite:6]index=6

XXXVI. 5. WHAT MUST BE COMPUTED CONCRETELY NOW

The folded-Bragg notes localize the remaining nite audit to the primitive-cell Fourier data

$$\text{Ø}_e\ (G\ ); \qquad \text{Þ}_e^1\ (G\ ); \qquad \text{Þ}_e^2\ (G\ ); \qquad v_k:$$

The constant intervalley Bragg block and enriched transverse blocks are controlled by

$$M_0 = \text{Ø}_e\ (G\ )1_2; \qquad W\ = \text{Þ}_e\ (G\ )s\ :$$

The local Weyl data are then extracted from

$$K_i = @_{k_i} L_{\mathrm{act}}(k)\ _{k=k}\ ; \qquad P\ K_i P\ :$$

:contentReference[oaicite:7]index=7

XXXVII. 6. HONEST NEXT STEP

So the next nonredundant computation is not another symbolic reduction.
It is exactly this nite operator build:

$$H_e(k+p) \;!\; H_{\text{shell}}(p) \;!\; P \quad !\; K_i \;!\; P \quad K_i P \;!\; (A;B;C)$$

with

$$(\ _k;\ ;\ ) = (A;B;C):$$

XXXVIII. EXPLICIT 4 4 FOLDED-BRAGG SHELL OPERATOR

XXXIX. 1. SHELL BASIS

Work in the folded-Bragg shell basis

$$H = \text{spanf}\ _{+;1}\ ;\ _{+;2}\ ;\ _{;1}\ ;\ _{;2}\ g;$$

with

$$_{+;a}(x) = e^{+iG\ x=2}u_a; \qquad _{;a}(x) = e^{\ iG\ x=2}u_a; \qquad a = 1;2:$$

In this basis, the shell operator is a 4 4 block matrix of the form

$$H_{\text{shell}}(p) = \begin{pmatrix} E^e_+(p) & B^e(p) \\ B^e(p)^y & E^e(p) \end{pmatrix};$$

where each block is 2 2.

XL. 2. FIRST-ORDER EXPANSION NEAR p = 0

Expand the valley-diagonal blocks as

$$E^e(p) = E\ \ 1_2\ \ v_k p_k 1_2 + O(jpj^{\ 2});$$

where v_k is extracted from the p_k-derivative of the valley-diagonal di erence. The intervalley block expands as

$$B^e(p) = M_{\ 0} + p_1 W^1 + p_2 W^2 + O(jpj^{\ 2});$$

with

$$M_0 = B^{\ e}(E\ ;0); \qquad W^i = @_{p_i} B^{\ e}(E\ ;p)_{\ p=0}:$$

Therefore the explicit rst-order shell operator is

$$H_{\text{shell}}(p) = E\ \ 1_4 + v_k p_k\ \ ^3\ \ 1_{\ 2} + \ \ ^1\ \ M_{\ 0} + p_1\ \ ^1\ \ W^{\ 1} + p_2\ \ ^1\ \ W^{\ 2} + O(jpj^{\ 2}):$$

XLI. 3. TRANSVERSE SHELL MATRICES

The transverse shell matrices are

$$W^1 = \hbar v_e^1\ (G\ )s^1; \qquad W^2 = \hbar v_e^2\ (G\ )s^2:$$

So the rst-order operator becomes

$$H_{\text{shell}}(p) = E\ \ 1_4 + v_k p_k\ \ ^3\ \ 1_{\ 2} + \ \ ^1\ \ M_{\ 0} + p_1\ \hbar v_e^1\ (G\ )\ \ ^1\ \ s^{\ 1} + p_2\ \hbar v_e^2\ (G\ )\ \ ^1\ \ s^{\ 2} + O(jpj^{\ 2}):$$

XLII. 4. MATRIX FORM

Writing

$$M_0 = \begin{pmatrix} m_{11} & m_{12} \\ \overline{m_{12}} & m_{22} \end{pmatrix}; \qquad s^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}; \qquad s^2 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix};$$

the shell operator is explicitly

$$\boxed{H_{\text{shell}}(p) = \begin{pmatrix} E + v_k p_k & 0 & m_{11} & m_{12} + \hbar p_e^1 p_1 - i\hbar p_e^2 p_2 \\ 0 & E + v_k p_k & \overline{m_{12}} + \hbar p_e^1 p_1 + i\hbar p_e^2 p_2 & m_{22} \\ m_{11} & \overline{m_{12}} + \hbar p_e^1 p_1 - i\hbar p_e^2 p_2 & E - v_k p_k & 0 \\ m_{12} + \hbar p_e^1 p_1 + i\hbar p_e^2 p_2 & m_{22} & 0 & E - v_k p_k \end{pmatrix} + O(|p|^2).}$$

Here $\hbar p_e = \hbar p_e(G)$.

XLIII. 5. PRINCIPAL SYMBOL AFTER THE SCALAR BRAGG SLOT IS REMOVED

On the graded bypass / carrier block, the constant scalar Bragg block is discarded, so the principal first-order shell symbol is

$$\boxed{A_1(p) = v_k p_k \, \sigma^3 \otimes \mathbb{1}_2 + p_1 \hbar p_e^1(G) \otimes s^1 + p_2 \hbar p_e^2(G) \otimes s^2.}$$

If the transverse amplitudes are equal after calibration,

$$\hbar p_e^1(G) = \hbar p_e^2(G) =: v_\perp,$$

then

$$\boxed{A_1(p) = v_k p_k \, \sigma^3 \otimes \mathbb{1}_2 + v_\perp \, p_1 \, \sigma^1 \otimes s^1 + p_2 \, \sigma^1 \otimes s^2.}$$

XLIV. 6. FINAL PROJECTED READOUT

Let P be the spectral projector onto the isolated two-dimensional low valley block. Define

$$K_i := \partial_{p_i} H_{\text{shell}}(p)\big|_{p=0}.$$

Then

$$\boxed{\gamma_k = \frac{1}{2}\operatorname{Tr}(\sigma^3 P K_k P); \qquad K_k = \hat{n}_i K_i;}$$

$$\boxed{= \frac{1}{4}\sum_{a=1}^{2} \operatorname{Tr}(\sigma^a P (e_{ai} K_i) P);}$$

$$\boxed{= \frac{1}{4}\Big[\operatorname{Tr}(\sigma^2 P (e_{1i} K_i) P)\ \operatorname{Tr}(\sigma^1 P (e_{2i} K_i) P)\Big].}$$

Equivalently, in the minimal axial ansatz

$$A_1(p) = A\, p_k \sigma^3 + B(p_1 \sigma^1 + p_2 \sigma^2) + C(-p_2 \sigma^1 + p_1 \sigma^2);$$

one simply has

$$\boxed{(\gamma_k; \gamma_\perp; \gamma_\parallel) = (A; B; C).}$$